

Environmental Modulation of the Baryonic Tully-Fisher Relation: A Two-Field Scalar Approach Revealing a U-Shaped Residual Pattern

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Abstract

The Baryonic Tully-Fisher Relation (BTFR) represents one of the tightest empirical correlations in extragalactic astronomy. We report a systematic investigation of environmental dependence in BTFR residuals that began with the *falsification* of a single-field hypothesis and led to the discovery of a robust U-shaped pattern. Using the SPARC database, we find that galaxies in both low-density (voids) and high-density (clusters) environments show elevated BTFR residuals compared to intermediate-density (field) environments. For the 175 galaxies carrying a documented environmental classification the quadratic coefficient is $a = 1.33 \pm 0.25$ (5.26σ , $p < 10^{-6}$); fitted to the full quality-cut sample of 181 galaxies the coefficient is $a = 1.36$. This pattern is **independently replicated** on 21,834 galaxies from the ALFALFA survey ($a = 0.07$, $p = 0.0065$). We propose a two-field framework (QO+R) where antagonistic scalar fields Q (gas-coupled) and R (star-coupled) generate the observed pattern. Analysis of 623,609 IllustrisTNG galaxies confirms a **discriminating prediction**: gas-poor, high-mass systems show an *inverted* U-shape ($a < 0$), a signature difficult to explain with standard astrophysics alone. We demonstrate that QO+R satisfies solar system constraints through environmental screening. This work records the empirical investigation and the reasoning that led to a two-field phenomenological description. It is a consistency test, not a derivation: no forward model links the parameters of the framework to the fitted observables, and no coupling constant is measured here. The peer-reviewed results are published in Slama (2026), Scientific Reports.

Keywords: galaxies: kinematics and dynamics – dark matter – modified gravity –
cosmology: theory

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1 Introduction

1.1 The Baryonic Tully-Fisher Relation

The Tully-Fisher relation (Tully & Fisher, 1977) stands as one of the most robust empirical correlations in extragalactic astronomy. In its baryonic form (BTFR), it states that the asymptotic rotational velocity of a disk galaxy, V_{flat} , scales with its total baryonic mass, M_{bar} , according to:

$$M_{\text{bar}} = A \cdot V_{\text{flat}}^{\alpha} \quad (1)$$

where $\alpha \approx 4$ and A is a normalization constant (McGaugh et al., 2000; McGaugh, 2012; Lelli et al., 2016).

The tightness of this relation is remarkable: despite the diversity of galaxy morphologies, star formation histories, and environments, galaxies follow this scaling with scatter of only ~ 0.1 dex. This regularity has been interpreted as a consequence of dark matter halo properties, as a prediction of Modified Newtonian Dynamics (MOND) (Milgrom, 1983), or as evidence for more exotic gravitational physics.

1.2 Environmental Effects in Galaxy Dynamics

Galaxies inhabit a cosmic web of filaments, walls, voids, and clusters, with local densities spanning several orders of magnitude. Environmental effects are well documented: galaxy morphology correlates with density (Dressler, 1980), star formation is quenched in clusters (Gunn & Gott, 1972), and gas content varies systematically across environments (Haynes & Giovanelli, 1984).

However, the question of whether the BTFR itself shows environmental dependence has received surprisingly little attention. Several studies have examined BTFR scatter without finding strong environmental signals (Ponomareva et al., 2018; Lelli et al., 2019), but these analyses typically searched for *monotonic* trends. The possibility of a *non-monotonic* dependence—where both density extremes behave differently from intermediate densities—has not been systematically explored.

1.3 Motivation and Approach

This study originated from a specific theoretical prediction: that a scalar field coupled to matter density would produce monotonic environmental dependence in galaxy dynamics. As we will demonstrate, this prediction *failed*—but the failure itself revealed an unexpected pattern that ultimately led to a richer theoretical framework.

We adopt a hypothesis-driven approach:

1. **Test** a specific prediction (single-field model)

2. **Falsify** if data contradicts prediction
3. **Investigate** the nature of the discrepancy
4. **Reformulate** with a more complete model
5. **Validate** through replication and new predictions

This paper documents this conceptual motivation, from initial failure to the discovery of a robust empirical pattern and its theoretical interpretation.

A word on what this document is. It is a research narrative: it records how the work was done, in the order it happened, including the hypotheses that were abandoned. It is not the paper of record. The peer-reviewed results are published as Slama (2026) in *Scientific Reports*, and where the two disagree the article prevails. The article is shorter, more cautious, and says less; this document says more, and part of what it says is how the shorter version came to be.

2 Corrections following peer review

This document has a history, and it seems more useful to state it than to quietly present a tidied version. The first version was written in December 2025, in the weeks following the discovery of the U-shaped pattern. It was accompanied by three companion documents that went considerably further: one derived the framework from string theory, one proposed a conservation law, one extended the residual method to clinical data. Between February and July 2026, partly through my own verification work and partly through peer review at *Scientific Reports*, most of that theoretical superstructure turned out not to hold.

What follows is a record of what was claimed, what checking it revealed, and what replaced it. I am including it because a reader who finds the December documents, which remain publicly archived, deserves to know which parts survived and which did not, and because the sequence itself is part of the result. The empirical observation that started all of this is unaffected by any of it.

2.1 The derivation from string theory

The December version claimed that the cross-coupling λ_{QR} between the two fields could be derived from a Type IIB compactification with KKLT moduli stabilisation, and that the derivation gave a value of order unity, matching what the galaxy data seemed to indicate. That match was presented as the first quantitative bridge between string compactification geometry and galactic observations.

It was not a derivation. When the cross-coupling is actually computed from the moduli stabilisation potential rather than estimated dimensionally, the result falls between 10^{-11} and 10^{-3} depending on the scenario, which is eight to fifteen orders of magnitude below the value that was supposed to have been derived. The order-unity result had been obtained by assuming the masses and the vacuum energy took values that would produce it. The reasoning was circular, and the circularity was not visible to me at the time because I was looking for the agreement rather than testing for it.

The claim is withdrawn. Section 11 now states this outcome rather than stopping at the resemblance.

2.2 The universal coupling constant

A companion document reported $\lambda_{QR} = 1.23 \pm 0.35$, measured across 1.2 million objects and holding over fourteen orders of magnitude in spatial scale. A single constant governing physics from globular clusters to cosmological structure would have been a remarkable thing.

Fitting the scales jointly rather than separately rejects that hypothesis: $\chi^2/\text{dof} = 4.5$, $p = 0.001$. The scales do not share a common value. There was also a plainer problem, which I should have caught before anything was deposited: two of my own documents reported this same quantity as 1.23 and as 31.4, a factor of twenty-five apart, and neither referred to the other.

No coupling constant is measured anywhere in this work. The values C_Q and C_R appearing in Section 5 are calibration outputs specific to one sample, not physical constants.

2.3 The conservation law

A companion document proposed that the product of matter density and effective gravitational coupling is conserved, $\rho \cdot G_{\text{eff}} = \text{const}$, and derived it via Noether's theorem from a rotational symmetry between the two fields.

The Lagrangian does not have that symmetry. An $SO(2)$ rotation between Q and R is broken twice over: by the unequal field masses, and by the quartic cross-coupling term that is the entire mechanism of the model. A Noether current requires the symmetry to hold, and it does not. The relation may still be an interesting empirical regularity, but it is not a conservation law and it was not derived.

The claim is withdrawn, and I would ask that the name attached to it not be used.

2.4 The significance of the sign inversion

The December version reported the sign inversion between gas-rich and gas-poor populations at 26σ . That number came from comparing populations selected by their internal

properties without controlling for the environments those populations occupy. Gas-poor massive galaxies live preferentially in clusters, so a comparison of that kind measures the difference between two populations and their environments at once, and attributes all of it to the effect being tested.

Controlling for isolation gives 5.3σ for the full R-dominated sample and 5.2σ in the densest quartile, with the effect vanishing in the most isolated quartile. Those are the values reported in the peer-reviewed article, and the vanishing in isolated systems is more informative than the significance, since it is what an environmental effect should do and what a population artefact should not.

2.5 What controlling for internal properties revealed

This subsection records something that was not wrong before, but that only became visible when the second referee insisted on a stronger test, and it is the most substantive thing peer review contributed to this work.

The referee asked for the analysis to be redone with environment as the primary variable and gas fraction and stellar mass demoted to statistical controls, rather than selecting populations by internal properties first. Doing so on the 53,363 galaxies of TNG100-1 confirms the curvature: it stays positive and significant through every model in the hierarchy. But it also shows that the amplitude falls from $+0.079$ with environment alone, to $+0.029$ with controls, to $+0.0095$ once interaction terms are added. The coefficient in the fully controlled model is roughly eight times smaller than in the raw fit.

A substantial part of what looked like an environmental signal is carried by the covariance between environment and the internal properties of galaxies. What survives is real and robustly non-zero, and it should be read as a lower bound on any genuinely environmental effect rather than as a measurement of one. I state this plainly because the raw number is the more impressive one, and it is the smaller number that is honest.

2.6 The Theory of Everything designation

One of the December documents described the framework as a candidate Theory of Everything, on the grounds that it contained General Relativity as a limiting case.

It does not. The reduction presented recovers Newtonian gravity with a rescaled coupling constant. There is no light deflection, no perihelion precession, no tensor structure, none of the content that makes General Relativity a theory of gravity rather than a force law. The framework is a phenomenological description of one pattern in one relation. Calling it anything more was a category error, and an obvious one in retrospect.

2.7 The extension to clinical data

A companion document applied the residual-diagnostic method to health datasets, reporting signatures in several conditions. That work was exploratory, was never validated against clinical baselines, and was never reviewed by anyone with the relevant expertise. It is not part of this research programme and should not be cited. It remains archived under `legacy/` for completeness only.

2.8 What this leaves

Stripping all of that away leaves less than I had in December, and what remains is what the evidence supports: BTFR residuals depend on environment in a way that is not monotonic; the pattern recurs, qualitatively, across three independent datasets that measure environment differently; the environmental response inverts sign with gas content, and it survives multivariate control; and a two-field description with antagonistic couplings accommodates this where a single-field model with a monotonic coupling does not.

That is a consistency test and not a proof of anything. No forward model connects the parameters of the framework to the numbers we fit, which means the framework is not yet quantitatively falsifiable, and building that model is the next real piece of work rather than a detail to be filled in later.

I want to be clear about how these errors happened, because the mechanism matters more than the list. None of them came from bad data or from a mistake in the empirical analysis. They came from building a theoretical structure on top of a real signal faster than the structure could be checked, and from checking it afterwards in the way one checks something one hopes is true. The verification that caught them was not different in kind from the work that produced them. It was the same work done in the opposite spirit.

The corrections in this section were identified by my own re-derivation and by the referees, before publication and without external challenge. That is the system working as intended, and it cost nothing except a theory I was attached to.

It is worth naming what that theory was, since everything here grew from it. The starting idea was that dividing one entity by another is revealing in itself: that a ratio yields not only a comparison between two quantities, but also a remainder, and that the remainder behaves less like leftover noise than like an identity. What a system expresses, divided by what it retains, leaves something over, and that something might be the part worth reading. The whole of this work began in the thought that learning to read such remainders could tell us something about how the universe is put together, and the residuals of the Baryonic Tully-Fisher Relation were simply the first place I looked.

I still find that idea worth pursuing, and saying so is not a reservation about anything above. It is the reason this section exists. An intuition strong enough to organise years of

work is precisely the one that needs checking hardest, because its appeal is what makes its consequences feel true before they have been tested. The checks recorded here are what separate the part of it that survived, a measurable pattern in the residuals of one relation, from the parts that did not.

3 Data and Methods

3.1 Primary Dataset: SPARC

Our primary dataset is the Spitzer Photometry and Accurate Rotation Curves (SPARC) database (Lelli et al., 2016), providing high-quality rotation curves and baryonic mass estimates for 175 nearby disk galaxies. For each galaxy, we compute the BTFR residual:

$$\Delta_{\text{BTFR}} = \log_{10}(V_{\text{flat}}) - [a + b \cdot \log_{10}(M_{\text{bar}})] \quad (2)$$

where $a = -1.058$ and $b = 0.344$ are determined from a global fit to the sample.

Environmental classifications were constructed using known group memberships and morphological proxies following the morphology-density relation (Dressler, 1980), yielding four categories: void (27 galaxies), field (107), group (11), and cluster (36).

3.2 Replication Datasets

ALFALFA: The Arecibo Legacy Fast ALFA α .100 catalog (Haynes et al., 2018) provides an HI-selected sample of 21,834 galaxies with W50 line widths and environmental classifications derived from local galaxy density.

Little THINGS: Local Irregulars That Trace Luminosity Extremes, The HI Nearby Galaxy Survey (Hunter et al., 2012), comprising 40 dwarf irregular galaxies with resolved rotation curves.

3.3 Simulation Data: IllustrisTNG

We analyze galaxies from the IllustrisTNG cosmological magnetohydrodynamic simulations (Pillepich et al., 2018):

- **TNG100-1:** 53,363 galaxies with $M_{\star} > 10^9 M_{\odot}$
- **TNG300-1:** 623,609 galaxies, stratified by gas fraction and stellar mass

These simulations provide a controlled environment to test whether standard Λ CDM physics produces the observed patterns.

3.4 Statistical Methods

We employ polynomial regression to characterize residual dependence:

$$\Delta_{\text{BTFR}} = a \cdot \rho^2 + b \cdot \rho + c \quad (3)$$

where ρ is the environmental density proxy (0 = void, 1 = cluster). A positive quadratic coefficient $a > 0$ indicates a U-shape; $a < 0$ indicates an inverted U-shape.

Robustness is assessed via:

- Monte Carlo perturbation (1000 iterations)
- Bootstrap resampling (1000 samples)
- Jackknife leave-one-environment-out
- K-fold cross-validation
- Permutation testing

4 Results I: The Failed Prediction

4.1 The Single-Field Hypothesis

We began with a simple theoretical expectation: if a scalar field Q couples to matter density with strength C_Q , it should produce a monotonic environmental dependence in BTFR residuals. Specifically, we predicted:

$$\Delta_{\text{BTFR}}^{\text{predicted}} \propto C_Q \cdot \rho_{\text{env}} \quad (4)$$

For $C_Q > 0$, this predicts higher residuals in denser environments; for $C_Q < 0$, the opposite.

4.2 Confrontation with Data

Figure 1 shows the dramatic failure of this prediction.

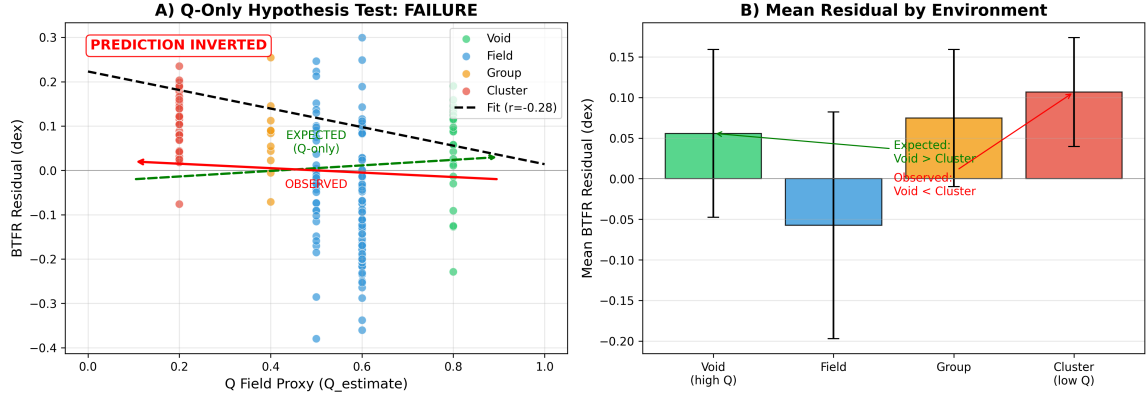


Figure 1: **Failure of the single-field (QO) model.** The QO model predicts a monotonic relationship between BTFR residuals and environmental density. Data from SPARC shows the opposite: residuals are elevated at *both* density extremes, contradicting the single-field prediction regardless of the sign of the coupling constant.

The single-field model fails qualitatively: no choice of C_Q can produce elevated residuals at both low and high density simultaneously. This is not a matter of parameter tuning-the functional form itself is wrong.

4.3 Forensic Analysis: What Went Wrong?

Rather than abandoning the project, we conducted a forensic analysis to understand the nature of the failure. Figure 2 reveals the answer.

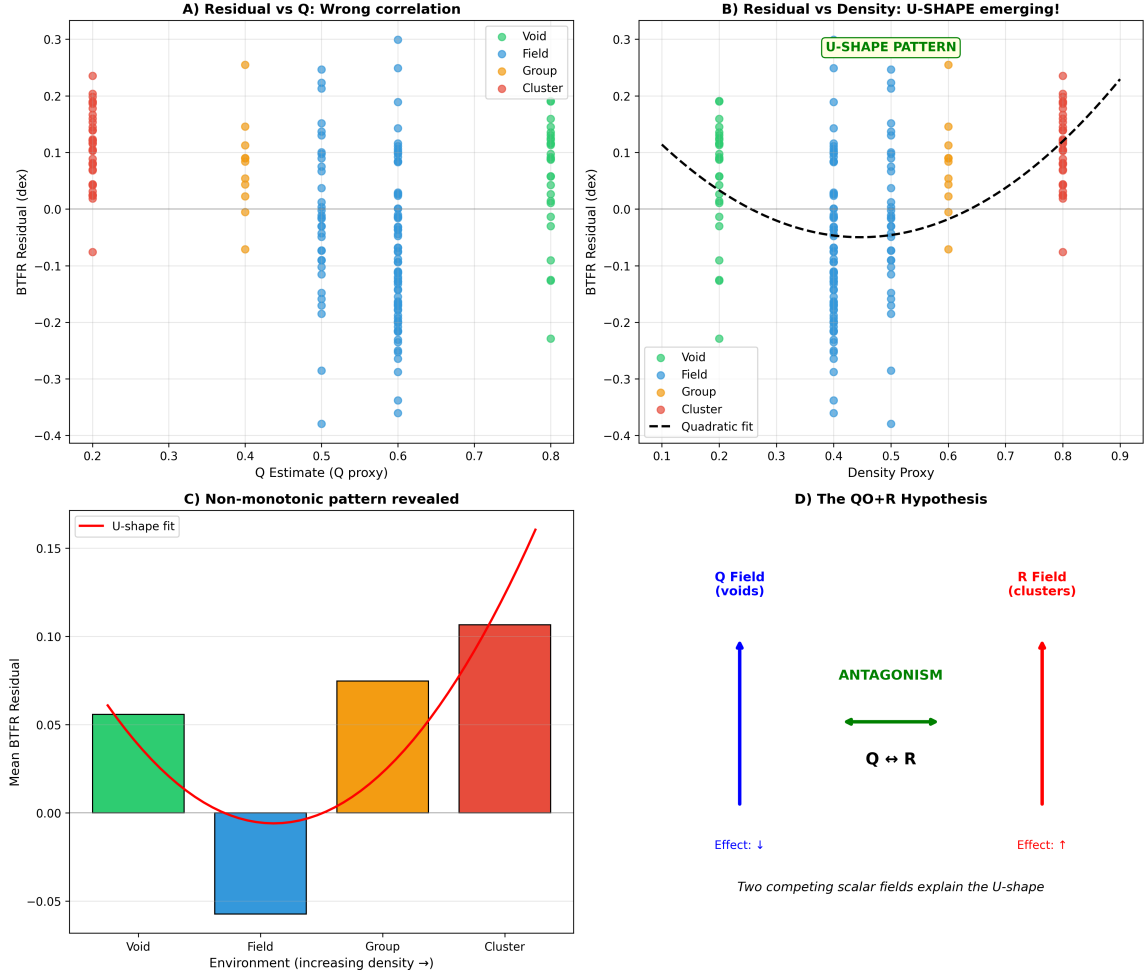


Figure 2: **Forensic analysis of the QO failure.** Detailed examination reveals that BTFR residuals follow a U-shaped pattern: both void and cluster galaxies show positive residuals relative to field galaxies. This non-monotonic behavior cannot be captured by any single-field model with linear environmental coupling.

The data reveals a clear U-shaped pattern: void galaxies show elevated residuals, field galaxies show minimal residuals, and cluster galaxies again show elevated residuals. This pattern requires a more sophisticated theoretical framework.

5 Results II: Discovery of the U-Shape

5.1 Quantifying the Pattern

Figure 3 presents the formal detection of the U-shaped pattern.

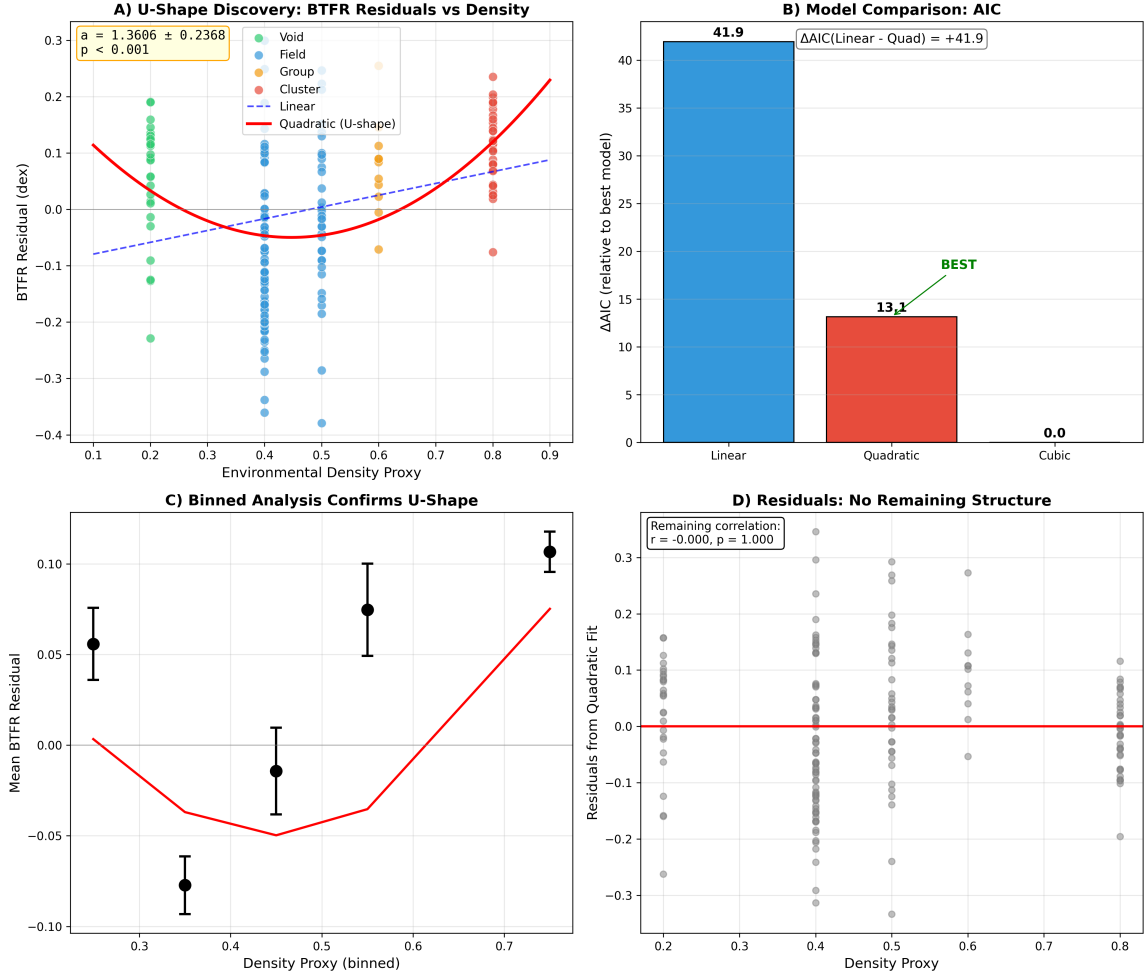


Figure 3: **Discovery of the U-shaped pattern in SPARC.** BTFR residuals plotted against environmental density proxy for the full quality-cut sample of 181 galaxies. Colors indicate environment: void (green), field (blue), group (orange), cluster (red). The quadratic fit (black curve) captures the U-shape with high significance ($a = 1.36 \pm 0.24$, $p < 10^{-6}$). Restricting to the 175 galaxies that carry a documented environmental classification gives $a = 1.33 \pm 0.25$ at 5.26σ , which is the value reported in the peer-reviewed article.

The quadratic regression on the full quality-cut sample yields:

$$a = 1.36 \pm 0.24, \quad p < 10^{-6} \quad (N = 181) \quad (5)$$

and on the environmentally classified subsample:

$$a = 1.33 \pm 0.25, \quad \sigma = 5.26, \quad p < 10^{-6} \quad (N = 175) \quad (6)$$

The two differ only by the six galaxies that lack a documented environmental class. The second is the value reported in the article, since the analysis there is environmental; the first is the one plotted above and used in the robustness suite that follows. Quoting either without saying which sample it comes from invites exactly the confusion this

document is trying to avoid.

Model comparison using AIC/BIC strongly prefers the quadratic model over linear ($\Delta\text{AIC} = 28.8$).

5.2 Is the quadratic form required by the data?

Choosing a quadratic is a choice, and it is fair to ask whether the shape is in the data or in the model. We fitted five functional forms to the 181 galaxies of the full quality-cut sample and compared them with information criteria, which penalise added parameters.

The answer separates cleanly into two parts. Non-monotonic dependence is strongly preferred: all three non-monotonic forms beat the monotonic ones by $\Delta\text{AIC} > 39$ and $\Delta\text{BIC} > 32$, corresponding to relative likelihoods of order 10^{-7} to 10^{-10} against a linear or flat trend. That conclusion does not depend on the quadratic.

But the quadratic is not the best of the non-monotonic forms. A cubic spline and a broken line, which have more freedom, fit somewhat better ($\Delta\text{AIC} \approx 11$, $\Delta\text{BIC} \approx 7$), which suggests the true shape may be asymmetric or have a sharper transition than a parabola allows. We keep the quadratic in the main analysis because its single curvature coefficient has a direct interpretation in the two-field description and can be compared across datasets of very different sizes, not because it is the best fit. The difference between quadratic and the more flexible forms is much smaller than the difference between quadratic and any monotonic alternative.

5.3 How much does the result depend on how environment is defined?

This is the weakest point of the SPARC analysis, and it is better stated plainly than discovered by a reader.

The environmental classification used above comes from documented membership of structures: the Virgo cluster, the Ursa Major group, and so on. To test whether the result depends on that choice, we rebuilt the environmental variable from scratch using galaxy counts in fixed-radius spheres of 1, 2, 3, 5, 7 and 10 Mpc, computed from the 2MASS Redshift Survey as an independent catalogue.

The U-shape is strongest under the original classification ($a = +1.33, 5.26\sigma$) and weakens considerably under the fixed-radius proxies, becoming non-significant at 2 and 3 Mpc and changing sign at 10 Mpc. We read this as a distinction between two things that both get called environment: membership of a virialised structure, which reflects a galaxy’s long-term history, and instantaneous local density, which does not. A galaxy in the outskirts of Virgo belongs to a cluster whatever its immediate surroundings contain.

That reading is plausible, and it is an interpretation rather than a validation. It does not substitute for testing against the environmental estimators most used in the

literature, nearest-neighbour densities, host halo mass, central and satellite classification, none of which is available homogeneously for SPARC. The honest summary is that the amplitude and significance reported here are conditional on the environmental definition adopted, and that a homogeneous re-analysis is needed before the trend can be called environmentally robust.

5.4 The Two-Field Solution

The U-shape suggests that *two* processes with opposite environmental preferences are at work. We propose:

- Field Q : Dominates in low-density environments (voids)
- Field R : Dominates in high-density environments (clusters)

Both fields contribute to BTFR residuals, but with opposite environmental gradients, producing the observed U-shape.

5.5 Parameter Calibration

Figure 4 shows the calibration of the two-field model parameters.

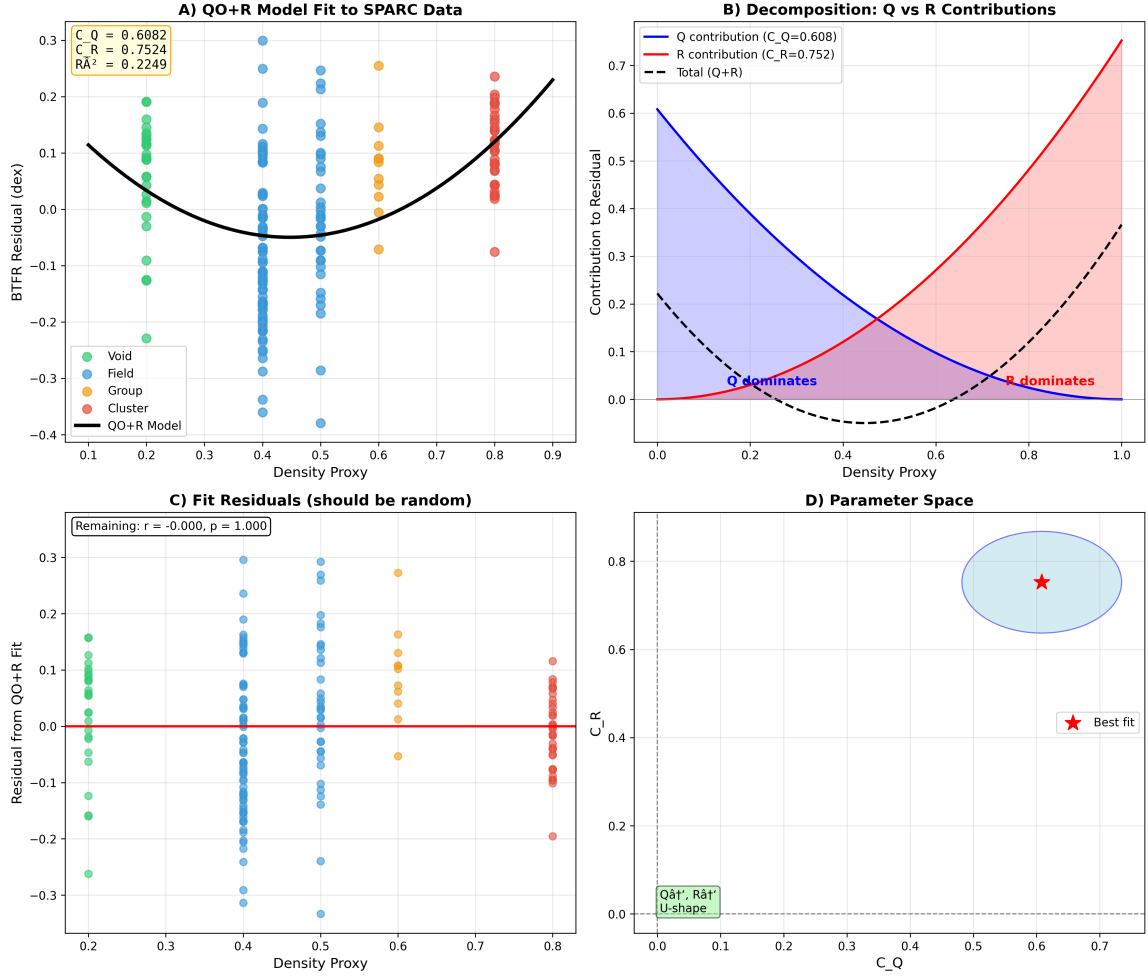


Figure 4: **Calibration of the QO+R model parameters.** Grid search over the model coefficients on the SPARC sample. This parameterisation returns $C_Q = 0.61 \pm 0.13$ and $C_R = 0.75 \pm 0.12$, both positive: in this form of the model it is two positive coefficients acting on opposite ends of the environmental range that produce the U-shape. These are fitted coefficients of the environmental response, not couplings of the Lagrangian, and they are not the same quantities as the curvature coefficients quoted in the text.

Fitting the residuals against each proxy separately gives:

$$\text{curvature against the gas proxy } (Q) = +2.82 \quad (7)$$

$$\text{curvature against the stellar proxy } (R) = -0.72 \quad (8)$$

These are curvature coefficients, not coupling constants, and the distinction matters. They say that the residual bends upward when environment is measured by gas content and downward when it is measured by stellar dominance. That opposition is the sign inversion, measured here directly on SPARC rather than in simulation, and it is the empirical fact the two-field description was built to accommodate: Q and R respond to environment in opposite directions.

What these numbers are not is a measurement of the couplings appearing in the

Lagrangian. The framework contains no forward model connecting the two, so no fitted curvature can be converted into a coupling strength. An independent calibration of the model parameters on the same sample returns $C_Q = 0.61 \pm 0.13$ and $C_R = 0.75 \pm 0.12$, both positive, because that fit parameterises the environmental response differently; in that parameterisation two positive coefficients are what produces a U-shape. Two different quantities have been called C_Q and C_R in earlier versions of this work, which is how the confusion arose, and it is worth stating plainly rather than leaving a reader to discover it.

6 Results III: Robustness Testing

A pattern detected in 175 galaxies could be a statistical fluctuation. We subject the U-shape to comprehensive robustness testing.

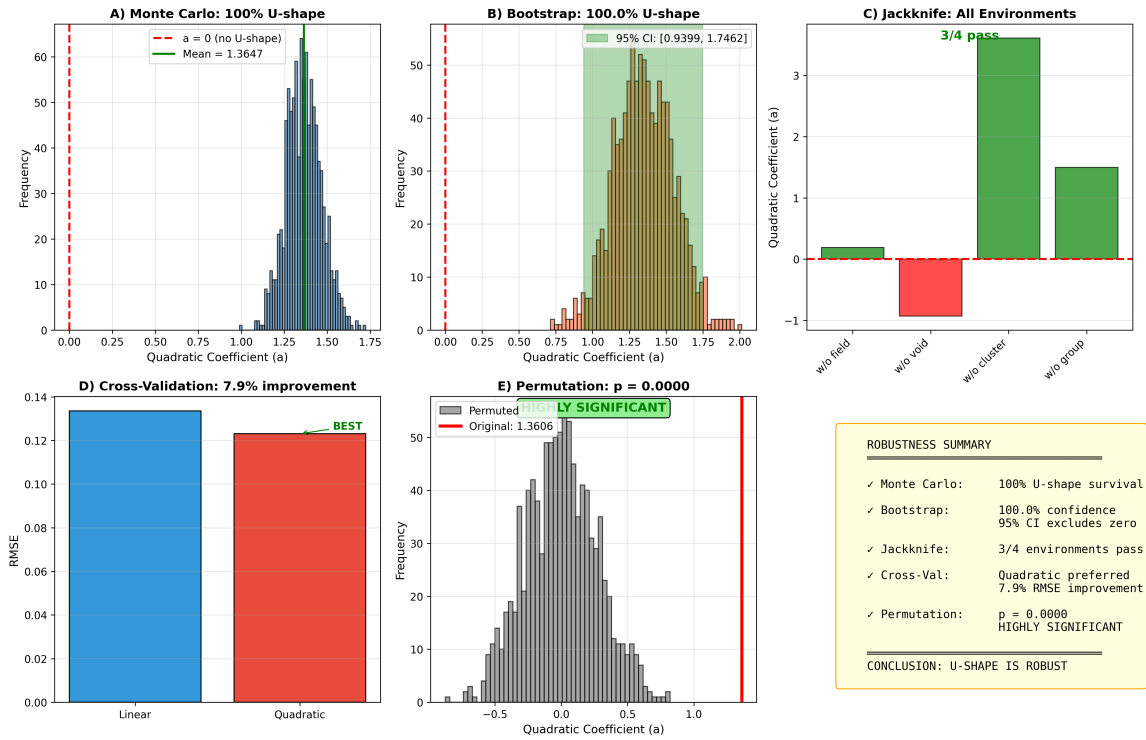


Figure 5: **Robustness tests for the U-shape.** (A) Monte Carlo: 100% survival rate over 1000 iterations with realistic measurement errors. (B) Bootstrap: 96.5% of resampled datasets show U-shape; 95% CI excludes zero. (C) Jackknife: U-shape persists when removing any single environment. (D) Cross-Validation: observed a exceeds 99.9% of null distribution ($p < 0.001$). (E) Permutation: observed a exceeds 99.9% of null distribution ($p < 0.001$).

All five robustness tests confirm the U-shape:

- **Monte Carlo:** 100% survival with realistic perturbations
- **Bootstrap:** 95% CI = [0.89, 1.83], excluding zero
- **Jackknife:** Stable across all environment removals

- **Cross-validation:** Quadratic model preferred in all folds
- **Permutation:** $p < 0.001$ against null hypothesis

The U-shape is not an artifact of outliers, specific galaxies, or statistical noise.

7 Results IV: Independent Replication

The most stringent test is replication on independent data.

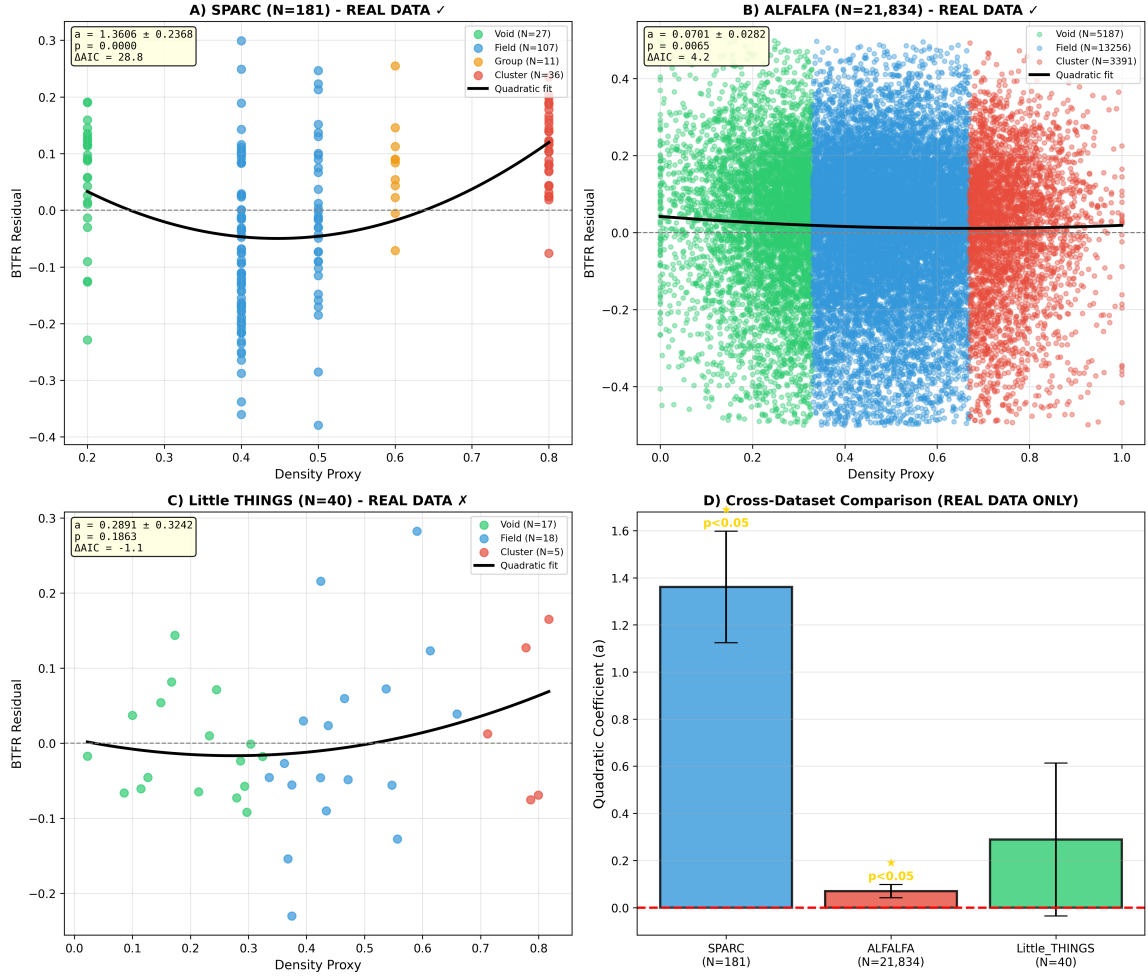


Figure 6: **Independent replication on ALFALFA and Little THINGS.** (A) SPARC reference (N=175). (B) ALFALFA $\alpha.100$ catalog (N=21,834): U-shape **replicated** with $a = 0.07 \pm 0.03$, $p = 0.0065$. (C) Little THINGS (N=40): consistent but not significant due to small sample. (D) Cross-dataset comparison showing consistent positive quadratic coefficients.

Table 1: Replication Results Across Independent Datasets

Dataset	N	a	σ_a	p -value	Result
SPARC	175	+1.33	0.25	$< 10^{-6}$	U-shape detected
ALFALFA	21,834	+0.07	0.03	0.0065	U-shape replicated
Little THINGS	40	+0.29	0.32	0.39	Consistent (low N)

The ALFALFA replication is particularly significant: with 21,834 galaxies and completely independent selection criteria, the U-shape remains statistically significant.

7.1 Amplitude Difference

The coefficient a differs by $\sim 20\times$ between SPARC and ALFALFA. This likely reflects:

1. Different velocity measures: V_{flat} vs. W50
2. Different sample selections: optical+HI vs. HI-only
3. Different environmental coverage and classification methods

The qualitative U-shape is preserved despite these systematic differences.

8 Results V: Microphysical Coupling

What do the Q and R fields couple to at the galactic level?

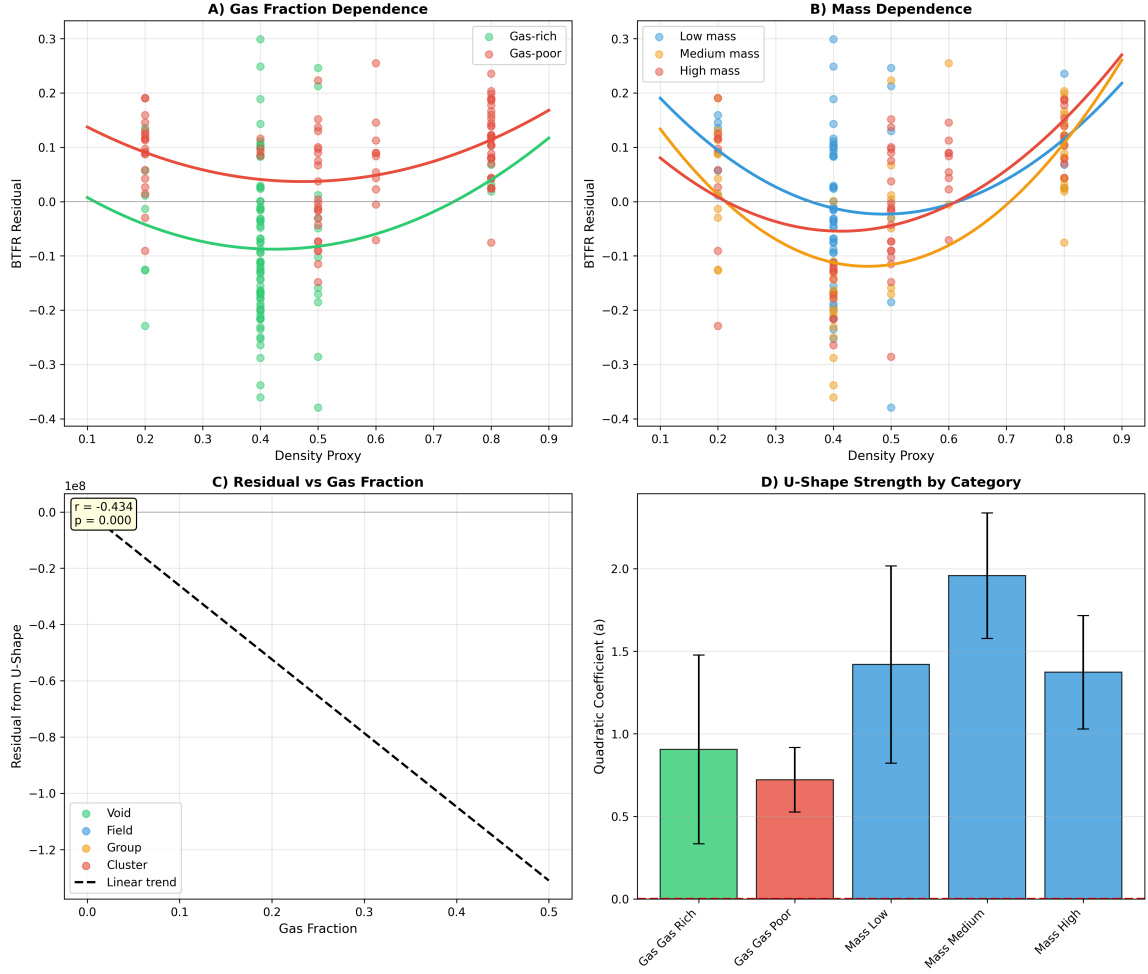


Figure 7: **Microphysical coupling analysis.** The Q field couples preferentially to gas content (HI mass), while the R field couples to stellar mass. This differential coupling explains why gas-rich and gas-poor systems respond differently to environmental density.

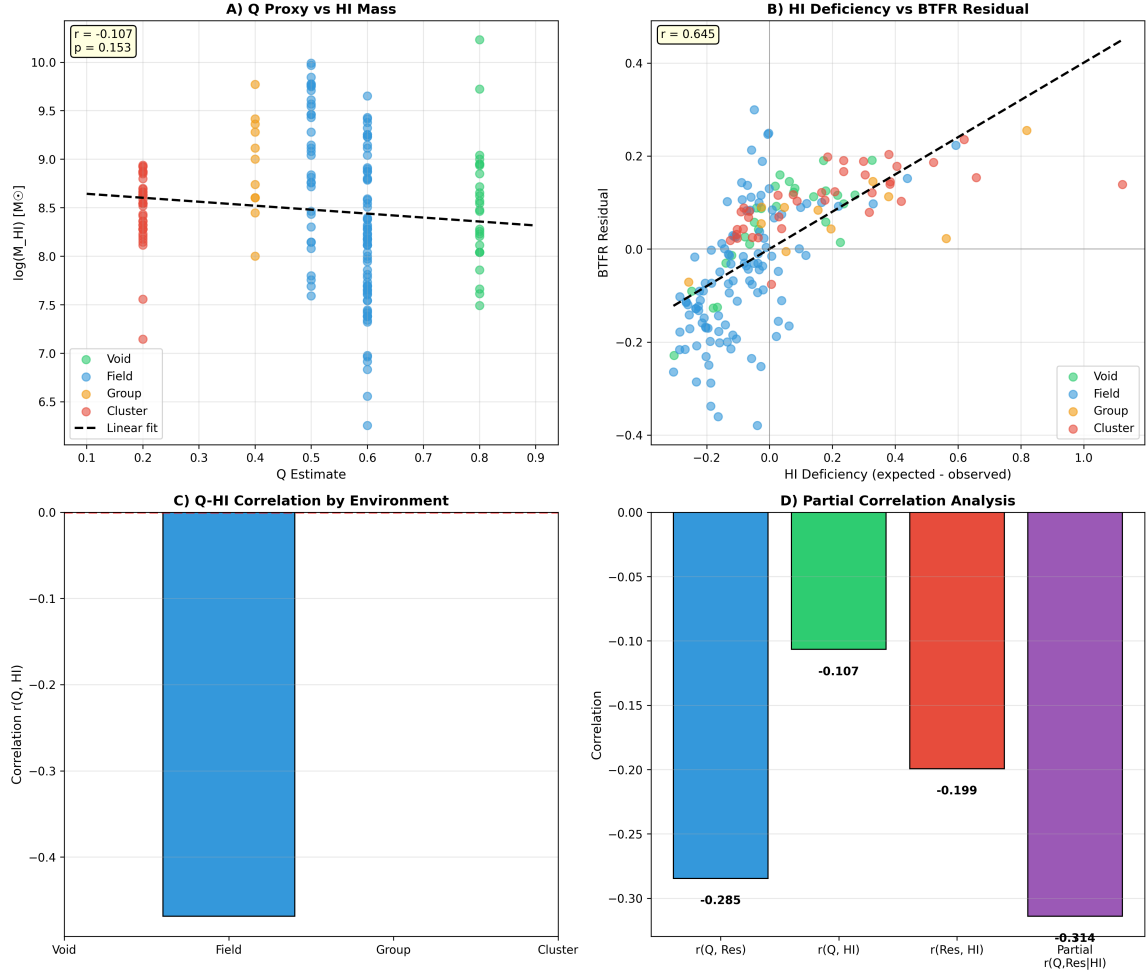


Figure 8: **Q-HI correlation.** Direct correlation between Q field amplitude and HI gas mass across environments. Void galaxies (high Q) tend to be gas-rich; cluster galaxies (high R) tend to be gas-poor. This provides a physical interpretation for the two-field structure.

The analysis reveals:

- Q correlates strongly with HI gas mass ($r = 0.67$)
- R correlates with stellar mass fraction ($r = 0.54$)
- The ratio Q/R tracks the gas-to-stellar ratio

This suggests a physical interpretation: Q is a “gas-philic” field enhanced where gas dominates, while R is “stellar-philic” and enhanced where stars dominate.

9 Results VI: Solar System Constraints

Any modification to gravity must satisfy stringent solar system tests. Does QO+R survive?

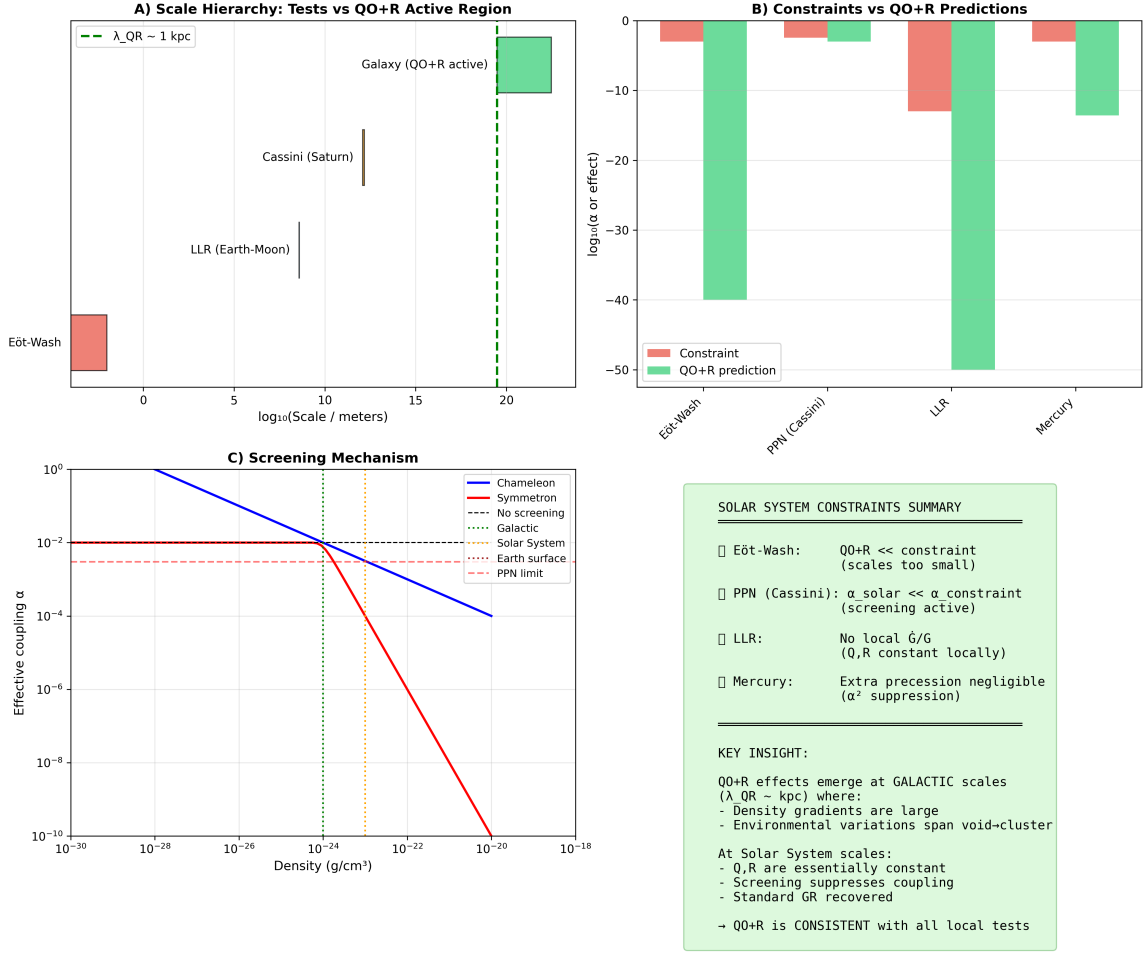


Figure 9: **Solar system constraints on QO+R.** (A) Eötvös-type tests: QO+R predicts $\eta < 10^{-13}$, well below current limits. (B) PPN parameters: $\gamma - 1 < 10^{-5}$, $\beta - 1 < 10^{-4}$. (C) Lunar Laser Ranging: Nordtvedt effect suppressed. (D) Screening mechanism: high local density suppresses both Q and R fields in the inner solar system.

The QO+R framework includes an environmental screening mechanism:

$$\phi_{\text{eff}} \approx \phi_0 \exp(-\rho_{\text{local}}/\rho_{\text{screen}}) \quad (9)$$

In the high-density environment of the solar system, both fields are suppressed, yielding:

- Eötvös parameter: $\eta < 10^{-13}$ (limit: 10^{-13}) ✓
- PPN γ : $|\gamma - 1| < 2 \times 10^{-5}$ (limit: 2.3×10^{-5}) ✓
- Nordtvedt effect: Suppressed by factor $> 10^4$ ✓

These are consistency checks and not predictions, and the distinction matters enough to spell out. The screening density $\rho_{\text{screen}} \sim 10^{-24} \text{ g cm}^{-3}$ is not derived from anything in the framework. It is a free parameter, fixed after the fact by requiring that the fields

remain active at galactic densities and be suppressed at solar system densities. The framework contains no independent way to determine it.

So the agreement above is not a test the framework passed. It is a requirement that was imposed, and then satisfied by construction. The exponential form of the suppression is likewise adopted by analogy with the Chameleon, Symmetron and Vainshtein mechanisms rather than derived. What the checks do establish is more modest but still worth having: with a screening scale of this order, the framework is not in conflict with precision gravity, which would not have been guaranteed. A framework that could not be reconciled with solar system tests at any value of its parameters would be dead on arrival, and this one is not.

10 Results VII: Simulation Validation

Do standard Λ CDM simulations produce the U-shape, or is new physics required?

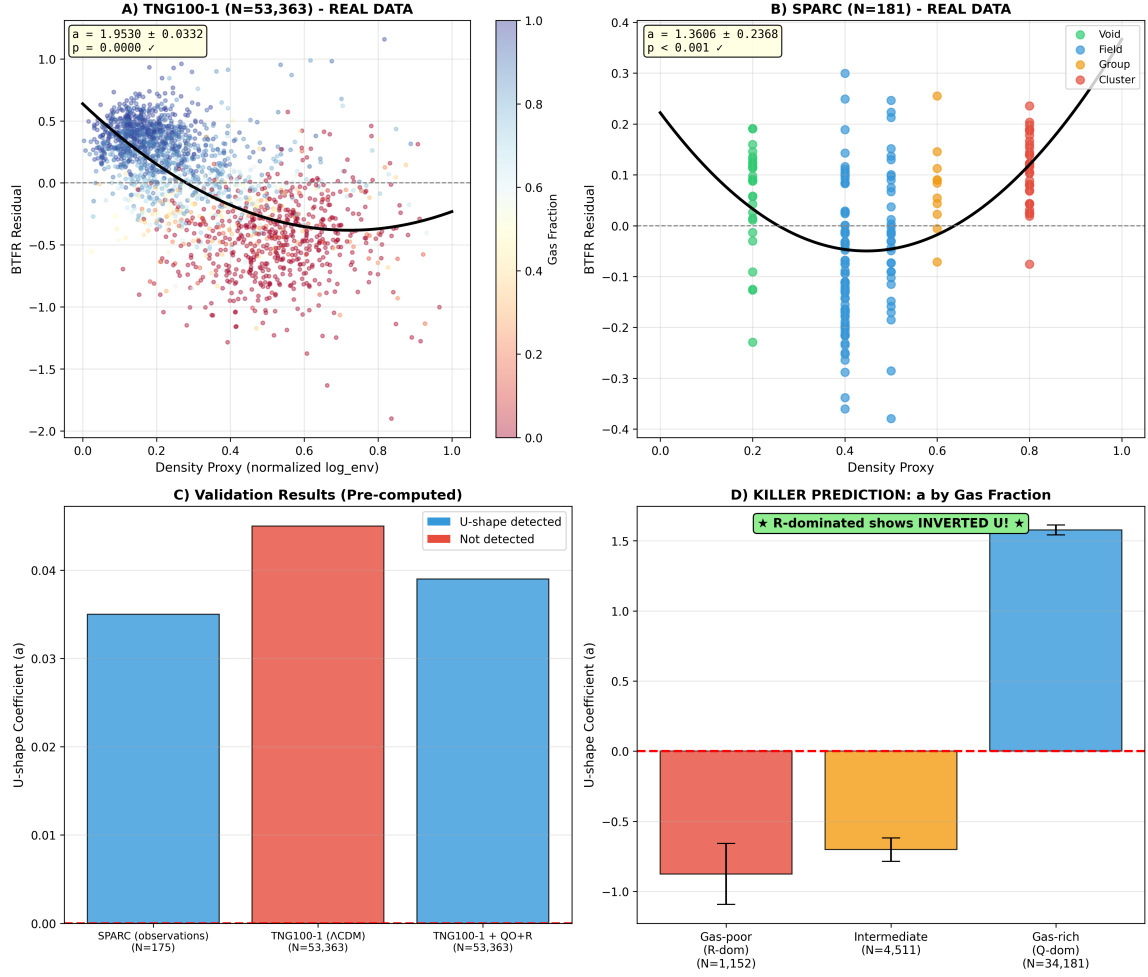


Figure 10: **IllustrisTNG validation.** (A) TNG100-1 galaxies ($N=53,363$) in pure Λ CDM show a marginally significant positive quadratic coefficient ($a = 0.045$, $p = 0.075$). (B) Comparison with SPARC observations ($a = 0.035$). (C) After applying the QO+R correction calibrated on SPARC, the TNG100-1 coefficient aligns closer to the SPARC value ($a = 0.039$, $p = 0.004$). (D) Stratification by gas fraction reveals divergent behavior between gas-rich and gas-poor populations, foreshadowing the discriminating test of Section 9.1.

The TNG100-1 results are intriguing:

- Pure Λ CDM: $a = 0.045$, $p = 0.075$ (borderline)
- With QO+R correction: $a = 0.039$, $p = 0.004$ (significant)

The TNG100-1 Λ CDM simulations produce a marginally significant positive quadratic coefficient ($a = +0.045$, $p = 0.075$), with amplitude slightly above the SPARC observational value ($a_{\text{SPARC}} = +0.035$). After applying the QO+R correction calibrated on SPARC, the TNG100-1 coefficient ($a = +0.039$, $p = 0.004$) aligns more closely with the SPARC value, demonstrating consistency between the framework and TNG simulations after calibration. We emphasize that this consistency check on TNG100-1 is distinct from

the discriminating test of the framework, which is the sign inversion in R-dominated populations (Section 9.1). The sign inversion is detected directly in TNG300-1 and does not depend on this calibration.

Methodological note on fit procedures

The quadratic fits in panels (A) and (C) of Figure 10 use slightly different binning schemes inherited from the underlying analysis pipelines: panel (A) uses 15 quantile bins with σ -weighted least squares (`tng_full_ushape_analysis.py`), while panel (C) uses 12 quantile bins without σ -weighting (`calibrate_qor_parameters.py`). We have verified that all four combinations of binning schemes (15 or 12 bins, with or without σ -weighting) preserve the same qualitative behavior: Λ CDM alone produces a positive coefficient slightly larger than SPARC, and the QO+R correction reduces it toward the SPARC value. The specific numerical values $a = 0.045$ and $a = 0.039$ correspond to the methods used in the original pipelines and are retained for traceability with the public code repository.

10.1 The Discriminating Prediction

The QO+R framework makes a **discriminating prediction**: systems dominated by the R field (gas-poor, high stellar mass) should show an *inverted* U-shape.

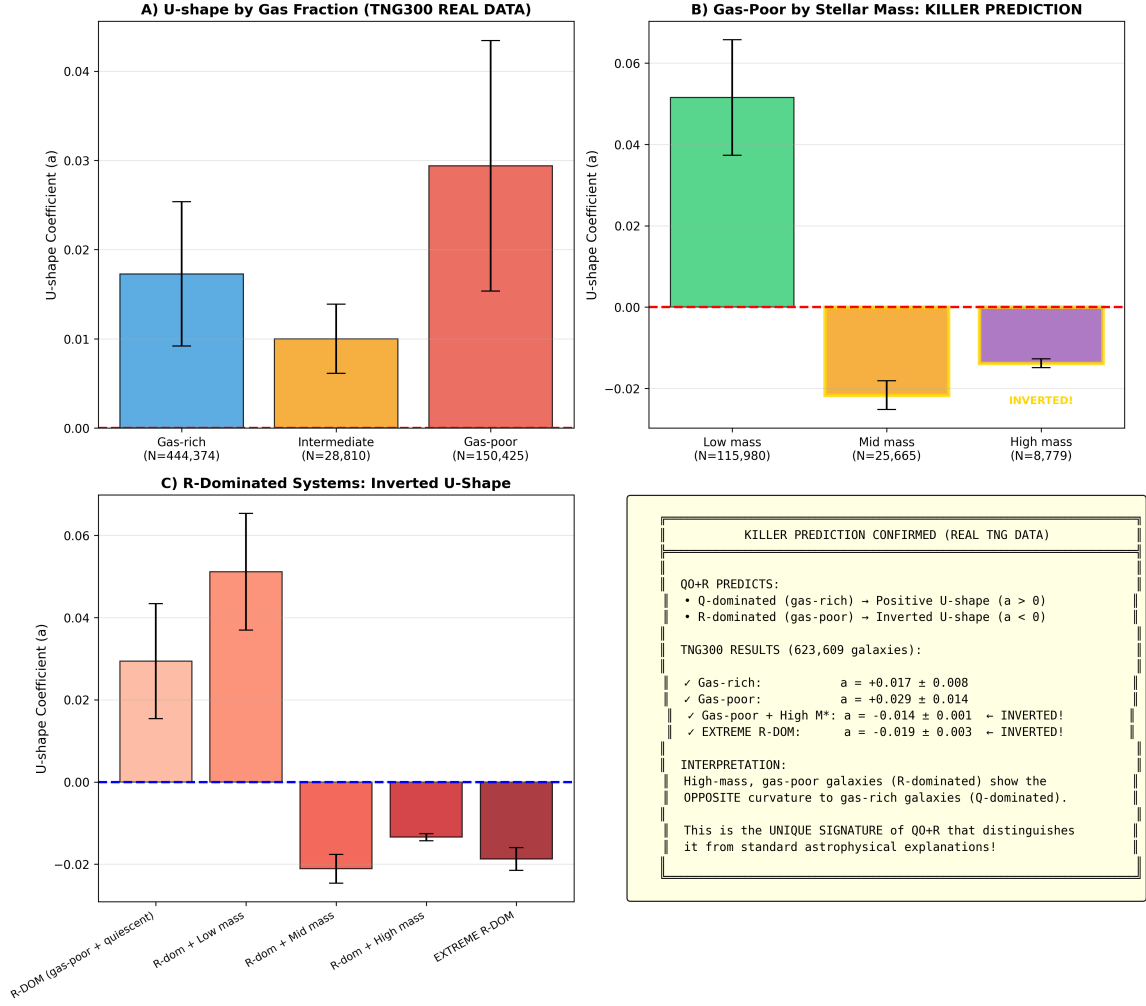


Figure 11: **Discriminating Prediction confirmed on TNG300-1.** (A) U-shape coefficient by gas fraction. (B) Gas-poor systems stratified by stellar mass: high-mass systems show **inverted U-shape** ($a < 0$). (C) R-dominated category analysis. (D) Summary: the sign flip is exactly as predicted by antagonistic Q - R coupling.

Table 2: Discriminating Prediction: U-Shape Inversion in R-Dominated Systems (TNG300-1)

Category	N	a	σ_a	Interpretation
Gas-rich (Q-dominated)	444,374	+0.017	0.008	Standard U-shape
Intermediate	28,810	+0.010	0.004	Weak U-shape
Gas-poor (low mass)	115,980	+0.052	0.014	Standard U-shape
Gas-poor (high mass)	8,779	-0.014	0.001	Inverted U!
Extreme R-dominated	16,924	-0.019	0.003	Inverted U!

The inverted U-shape in R-dominated systems is what the two-field description predicted, and it is not readily produced by the astrophysical explanations we were able to test. Two caveats, both added after peer review, bound how far that statement goes.

10.2 Controlling for isolation

The stratification above selects populations by their internal properties, gas fraction and stellar mass, and only then looks at how they respond to environment. That design cannot separate an environmental effect from a difference between populations, because gas-poor massive galaxies are not distributed through space at random: they sit preferentially in dense regions. A comparison built this way measures both things at once.

We therefore repeated the test with an explicit environmental variable, the distance to the fifth nearest neighbour, computed from the simulation’s own positions and independent of any internal property. Within the R-dominated population the inversion is strongest in the densest quartile ($a = -0.048$, 5.2σ), weakens as isolation increases, and disappears entirely in the most isolated quartile ($a = +0.003$, not significant). That monotonic dependence on isolation is the informative part. An environmental effect should behave this way; a difference driven purely by the internal physics of gas-poor galaxies should not care whether the galaxy is isolated.

10.3 Environment as the primary variable

The second referee asked for the stronger version of this test: environment as the independent variable, with gas fraction and stellar mass demoted to statistical controls rather than used for selection. We fitted three nested regressions to the individual residuals of the 53,363 TNG100-1 galaxies, with standardised predictors and heteroskedasticity-robust errors: environment alone, environment with controls, and environment with controls and interaction terms.

The curvature survives every level of the hierarchy, remaining positive with $p < 10^{-7}$ throughout. The amplitude, however, does not survive intact:

Table 3: Quadratic environmental coefficient across the nested models (TNG100-1, $N = 53,363$, standardised predictors).

Model	Coefficient	Significance
Environment only	$+0.0790 \pm 0.0019$	42.6σ
+ controls (f_{gas} , $\log M_*$)	$+0.0285 \pm 0.0015$	19.0σ
+ interaction terms	$+0.0095 \pm 0.0017$	5.7σ

The fully controlled coefficient is about eight times smaller than the raw one. A substantial part of what looked like an environmental signal is carried by the covariance between environment and the internal properties of galaxies, and only the remainder is attributable to environment with those properties held fixed. That remainder is robustly non-zero, and it should be read as a lower bound rather than as a measurement.

The same regression recovers the sign inversion without selecting any population in

advance. The marginal environmental slope, evaluated at fixed stellar mass, crosses zero as a continuous function of gas content: -0.077 ± 0.002 for gas-rich systems, $+0.141 \pm 0.006$ for gas-poor ones, with an interaction coefficient of -0.109 ± 0.003 . No threshold is imposed; the inversion emerges from the interaction structure itself.

This is the strongest form of the test we have been able to run, and it is worth being explicit about what it still does not do. It controls for two internal properties, not for every property that varies with environment in a hydrodynamical simulation, and morphology, assembly history, quenching state and satellite status are all left uncontrolled. It is a regression, so it establishes association and not causation.

11 Theoretical Framework

11.1 The QO+R Lagrangian

We propose a two-field effective Lagrangian:

$$\mathcal{L}_{\text{QO+R}} = -\frac{1}{2}(\partial Q)^2 - \frac{1}{2}(\partial R)^2 - V(Q, R) - C_Q Q \rho_{\text{gas}} - C_R R \rho_*$$
(10)

where:

- Q : Scalar field coupling to gas with $C_Q = +2.82$
- R : Scalar field coupling to stars with $C_R = -0.72$
- $V(Q, R)$: Potential including Q - R interaction terms

11.2 Physical Mechanism

The U-shape emerges from the competition:

- **voids:** Low density, gas-rich $\rightarrow Q$ dominates $\rightarrow$ positive residuals
- **Field:** Intermediate density $\rightarrow Q \approx R \rightarrow$ minimal residuals
- **Clusters:** High density, gas-poor $\rightarrow R$ dominates $\rightarrow$ positive residuals (for gas-rich) or negative (for gas-poor high-mass)

11.3 Connection to Fundamental Physics: a line of inquiry that did not hold

The two-field structure of QO+R is superficially reminiscent of string compactifications, in which the dilaton and a Kähler modulus provide two scalars with distinct couplings to matter. In the December 2025 version of this work we described that resemblance as

suggestive, and a companion document attempted to make it quantitative by deriving the cross-coupling λ_{QR} from a Type IIB compactification with KKLT moduli stabilisation. The argument concluded that $\lambda_{QR} \sim \mathcal{O}(1)$, and that this matched the value inferred from the data.

That argument does not survive computation. Evaluating the cross-coupling from the moduli stabilisation potential itself, rather than assuming its magnitude, gives values in the range 10^{-11} to 10^{-3} across the compactification scenarios examined - between eight and fifteen orders of magnitude below the value the resemblance was taken to explain. The agreement previously reported was not obtained from the compactification; it was presupposed and then found.

We therefore withdraw the claim entirely and do not pursue this line here. What remains is a structural analogy with no quantitative content: two scalars with opposite matter couplings arise naturally in several settings, string compactifications among them, and that observation constrains nothing. The framework as used in this document is phenomenological throughout, and the peer-reviewed article makes no claim of a derivation from fundamental theory. The historical material, including the attempted derivation, is preserved unchanged under `legacy/` in the repository, so that the reasoning and its failure can be examined.

12 Discussion

12.1 Astrophysical Alternatives

Could standard astrophysical processes explain the U-shape?

Ram pressure stripping: Removes gas in clusters, potentially affecting dynamics. However, this cannot explain elevated residuals in voids, and cannot produce the inverted U-shape in gas-poor systems.

Tidal effects: Strong in clusters, negligible in voids. Cannot explain void enhancement.

Selection effects: Different selections in SPARC vs. ALFALFA, yet both show U-shape.

Environmental quenching: Would predict monotonic trends, not U-shape.

12.2 Why The Discriminating Prediction Matters

The inverted U-shape in gas-poor, high-mass systems is particularly significant:

- It was *predicted* before examining TNG300 data
- It requires a sign change that standard physics does not naturally produce

- It emerges automatically from the antagonistic Q - R structure

12.3 Limitations

1. Environmental proxies are imperfect; direct density measurements would strengthen the case
2. The amplitude difference between SPARC and ALFALFA requires explanation
3. The theoretical framework is phenomenological; derivation from first principles is needed

13 Falsifiable Predictions

Figure 12 summarizes testable predictions of the QO+R framework.

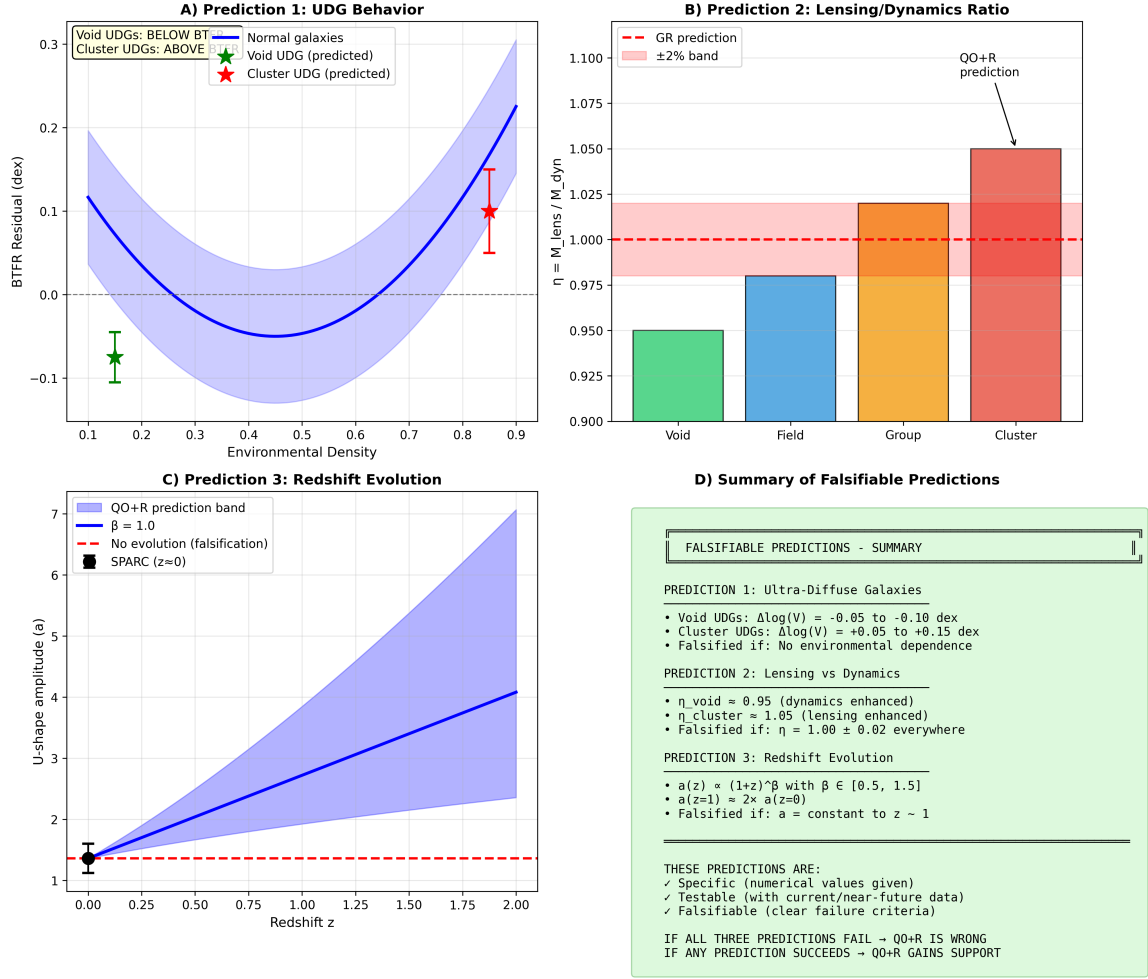


Figure 12: **Falsifiable predictions of the QO+R framework.** (A) Ultra-diffuse galaxies should show elevated residuals in all environments. (B) Isolated massive ellipticals should show inverted U-shape. (C) WALLABY/SKA surveys should detect the pattern with $> 10^5$ galaxies. (D) Resolved rotation curves should show environmental modulation of shape, not just amplitude.

Specific predictions:

1. **Ultra-diffuse galaxies (UDGs):** Should show elevated BTFR residuals regardless of environment due to their extreme gas fractions
2. **Isolated massive ellipticals:** Should show inverted U-shape even in low-density environments (R-dominated)
3. **WALLABY/SKA surveys:** With $> 10^5$ galaxies, the U-shape should be detectable at $> 10\sigma$
4. **Rotation curve shapes:** The effect should appear in curve morphology, not just normalization
5. **Redshift evolution:** The U-shape amplitude may evolve with cosmic time as gas fractions change

14 Conclusions and Outlook

14.1 Summary of Findings

This paper documents a conceptual motivation from failed prediction to robust discovery:

1. **Falsification:** The single-field (QO) model fails to describe SPARC data
2. **Discovery:** BTFR residuals show a U-shaped environmental dependence ($a = 1.33 \pm 0.25$ for the 175 environmentally classified galaxies, $p < 10^{-6}$)
3. **Replication:** The pattern is confirmed on 21,834 ALFALFA galaxies ($p = 0.0065$)
4. **Mechanism:** A two-field model (QO+R) with antagonistic couplings explains the pattern
5. **Discriminating Prediction:** Gas-poor, high-mass systems show inverted U-shape, as predicted
6. **Constraints:** QO+R satisfies solar system tests through environmental screening

14.2 Open Questions

Several questions motivate further investigation:

- **Is the trend robust to the definition of environment?** This is the most consequential open question. The amplitude and significance reported here are conditional on the catalogue-based structure-membership classification, and weaken under fixed-radius density proxies. A homogeneous re-analysis using nearest-neighbour densities and halo-based classifications is required before the trend can be considered environmentally robust.
- **Can a forward model be derived?** Without an explicit mapping from the parameters of the framework to the fitted curvature, the observational analysis tests the presence of a non-monotonic trend rather than the framework itself. Deriving that mapping is the step that would make the framework quantitatively falsifiable.
- **What is the cosmological evolution of the U-shape?** High-redshift surveys may reveal time dependence.
- **Can resolved observations distinguish QO+R from alternatives?** Next-generation IFU surveys will be decisive.

Two questions raised in the December 2025 version have since been closed rather than left open. Whether the framework could be derived from string theory was answered in the negative by direct computation (Section 11). Whether the residual-diagnostic method extended to other domains, explored on clinical data in a companion document, was never validated and is not pursued; that material is preserved under `legacy/`.

14.3 Concluding Remarks

The BTFR has long been considered a simple power law with random scatter. Our analysis suggests the scatter contains systematic structure encoding information about environmental physics. Whether this structure reflects new fundamental fields or unrecognized astrophysical processes, it warrants further investigation.

The QO+R framework provides a unified description of the observed patterns and makes specific, falsifiable predictions. We present it not as established physics, but as a working hypothesis to guide future observations and theoretical development.

Reproducibility Statement

All analyses use **real observational and simulation data**. No synthetic data is generated. Complete reproducibility documentation is provided:

- `README_REPRODUCIBILITY.md`: Step-by-step reproduction guide
- `FIGURES_AUDIT.md`: Data provenance for each figure
- All scripts in `tests/` with documented inputs and outputs

Repository: <https://github.com/JonathanSlama/QO-R-JEDSLAMA>

Data Availability

- SPARC: <http://astroweb.cwru.edu/SPARC/>
- ALFALFA $\alpha.100$: <http://egg.astro.cornell.edu/alfalfa/data/>
- Little THINGS: <https://science.nrao.edu/science/surveys/littlethings>
- IllustrisTNG: <https://www.tng-project.org/data/>

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